

PAPER_1592 — Bohr Magneton $\mu_B \approx \mu_B = K_MEX \cdot D_phys + SSQ + F \cdot D - F^2 \cdot D + F^2 = 9.274$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum/EM **Date:** June 18, 2026 **Status:** CLOSED — 0.007% residual near-EXACT closure

Observation

PAPER_1209DD_S622 (Quantum/EM Unified Proof Set) gives the lead-digit form:

Bohr Magneton $\mu_B \approx \mu_B = K_MEX \cdot D_phys + SSQ + F \cdot D - F^2 \cdot D + F^2 = 9.274$

Residual to CODATA: **0.007%**.

UQFF Closed Identity

$\mu_B = K_MEX \cdot D_phys + SSQ + F \cdot D - F^2 \cdot D + F^2 \approx 9.274$ residual 0.007%

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around **$K_MEX \cdot D_phys = 8.33$** or **$N_CH = 9$** - **Planck/quantum constants** ($\hbar$) cluster around **$D_BSFG = 6$** - **Relativistic constants** (c) cluster around **$SO_5/D_phys = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Bohr Magneton μ_B as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S622
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "bohr_magneton_mu_b_9_274"})`

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